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### Systems and components for regulating fluid infusion to a patient

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#### Abstract

Various fluid delivery systems are described comprising an infusion pump having a housing with a first opening and a hollow interior portion that is configured to receive a cartridge having a tubing. A pump unit can be disposed within the housing. The pump unit comprises a motor mechanically coupled with a crank shaft or eccentric cam that is configured to move a set of pistons or other objects to thereby compress one or more portions of the tubing over time as the crank shaft or eccentric cam rotates.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name      | U.S. Cl.  | CPC              |
|--------------|-------------|--------------------|-----------|------------------|
| 3985133      | 12/1975     | Jenkins et al.     | N/A       | N/A              |
| 4204538      | 12/1979     | Cannon             | N/A       | N/A              |
| 4207871      | 12/1979     | Jenkins            | N/A       | N/A              |
| 4314567      | 12/1981     | Cannon             | N/A       | N/A              |
| 4404440      | 12/1982     | Busche             | N/A       | N/A              |
| 4424720      | 12/1983     | Bucchianeri        | N/A       | N/A              |
| 4460358      | 12/1983     | Somerville et al.  | N/A       | N/A              |
| 4561830      | 12/1984     | Bradley            | N/A       | N/A              |
| 4617014      | 12/1985     | Cannon et al.      | N/A       | N/A              |
| 4690673      | 12/1986     | Bloomquist         | N/A       | N/A              |
| 5165874      | 12/1991     | Sancoff            | 417/474   | A61M<br>5/14228  |
| 5647854      | 12/1996     | Olsen et al.       | N/A       | N/A              |
| 5741121      | 12/1997     | O'Leary            | N/A       | N/A              |
| 5853386      | 12/1997     | Davis et al.       | N/A       | N/A              |
| 6146109      | 12/1999     | Davis et al.       | N/A       | N/A              |
| 7648627      | 12/2009     | Beden              | 210/85    | A61M<br>1/362223 |
| 8118778      | 12/2011     | Haylor et al.      | N/A       | N/A              |
| 11484645     | 12/2021     | Patton             | N/A       | F04B 43/025      |
| 2001/0004444 | 12/2000     | Haser              | 417/477.2 | A61M 3/0201      |
| 2004/0037723 | 12/2003     | Herwig et al.      | N/A       | N/A              |
| 2005/0118048 | 12/2004     | Traxinger          | 417/477.2 | A61M 1/72        |
| 2009/0043252 | 12/2008     | Haylor et al.      | N/A       | N/A              |
| 2011/0313358 | 12/2010     | Hariharesan et al. | N/A       | N/A              |
| 2012/0130309 | 12/2011     | Hariharesan        | 604/151   | A61M 39/281      |
| 2015/0073338 | 12/2014     | Waldhoff           | 604/67    | A61J 15/0026     |
| 2015/0182688 | 12/2014     | Dhami              | N/A       | N/A              |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 2012097141 | 12/2011          | WO      | N/A |

**OTHER PUBLICATIONS**

European Patent Office, International Search Report and Written Opinion, Mar. 15, 2022. cited by applicant

First Office Action for CN Application No. 202111370522.7 dated May 1, 2025, 4 pages (including English Translation). cited by applicant

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**Background/Summary**

(1) This application is a continuation application of U.S. non-provisional patent application having Ser. No. 17/529,691 filed on Nov. 18, 2021, which itself claims priority to U.S. provisional patent application having Ser. No. 63/115,443 filed on Nov. 18, 2020. This and all other referenced extrinsic materials are incorporated herein by reference in their entirety. Where a definition or use of a term in a reference that is incorporated by reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein is deemed to be controlling.

**FIELD OF THE INVENTION**

(1) The field of the invention is fluid infusion devices, and in particular, intravenous fluid infusion pumps.

**BACKGROUND**

(2) The following description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

(3) Various pumps exist for fluid infusion to a patient. One of the most common are gravity infusion devices, which utilize gravity to deliver medication. However, such devices are unable to provide precise dosage rates and any information concerning the fluid to be infused.

(4) In an attempt to address these deficiencies, various companies offer volumetric pumps that allow for more precise dosage rates. For example, ALARIS® offers its CAREFUSION® infusion pumps that can provide different dosage rates for multiple medications. Among other problems, such pumps are typically bulky, have a limited user interface, can have the door assembly open unexpectedly resulting in suspension of the medication to the patient, and can be difficult to prime and begin a line of infusion.

(5) All publications identified herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

(6) Thus, there is still a need for improved infusion devices that are easier to use and understand,

easier to prime, and provide additional safeguards to prevent unnecessary suspensions and other issues.

## SUMMARY OF THE INVENTION

(7) The inventive subject matter provides apparatus, systems and methods for intravenous (IV) infusion devices. Contemplated fluid delivery systems comprise a housing having a first opening leading into a hollow interior portion. Preferably, the first opening and hollow interior portion are configured to receive at least a portion of a cartridge having a tubing.

(8) It is contemplated that the cartridge comprises a housing that is configured to retain a portion of a tubing through which a medication can flow from a medication source to a patient.

(9) A pump unit is disposed within the housing and comprises a motor that is mechanically coupled with an eccentric cam via one or more gears, such that the motor causes rotation of the eccentric cam. Preferably, the eccentric cam is connected or coupled to one or more pistons that are configured to move as the eccentric cam rotates. When the cartridge is disposed within the hollow interior portion, the tubing of the cartridge is disposed adjacent or near the pump unit, such that rotation of the eccentric cam causes the pistons to depress different portions of the tubing over time, and thereby causes flow of a fluid within the tubing at a flow rate that is based on the rotation of the eccentric cam.

(10) Contemplated fluid delivery systems preferably also comprise a display communicatively coupled with the pump unit and configured to display information concerning a patient and a medication delivery, for example. Preferably, the display is a touch-screen display or otherwise configured to allow input of commands from a medical professional. For example, using the display, it is contemplated that a dosage of medication being delivered to a patient can be varied by the medical professional.

(11) Preferably, the display is physically separate from the housing and communicatively coupled with the pump unit via a wired or wireless connection. In such embodiments, it is preferred that the display can be attached or otherwise coupled to an IV pole and disposed to be approximately at an eye height level of a user. To allow for proper ergonomics, it is contemplated that the display could be vertically moved up and down the IV pole until a desired height is reached. The display could also be rotated about the IV pole. In some contemplated embodiments, the display could be tilted up and/or down to facilitate viewing of the information on the display.

(12) In some embodiments, the tubing may enter a top portion of the cartridge housing and exit through a bottom portion of the cartridge housing. It is preferred, though not necessary, that the tubing enters and exits through a front surface or face of the cartridge housing (e.g., the surface facing away from the interior portion of the cartridge). In some embodiments, a portion of the tubing that extends generally horizontally through the cartridge can be disposed at an angle greater than zero degrees from a horizontal surface, meaning that the tubing moves lower with respect to the cartridge housing as the tubing moves from the face to the back of the housing. This can be helpful to cause any air bubbles within the tubing to flow upwardly (e.g., toward the face of the housing) and out of the cartridge.

(13) The cartridge preferably comprises an actuator coupled to a valve that inhibits a flow of fluid in the tubing. The valve is configured to compress the tubing to prevent fluid flow in the tubing of the cartridge when the valve is in a first position. This advantageously prevents unintentional flow of fluid through the cartridge. Preferably, the valve is biased in the first position such as by using a spring to thereby prevent the flow of fluid when the actuator is not engaged.

(14) When the actuator is depressed or otherwise engaged, the valve is moved to a second position that allows fluid to flow through the tubing of the cartridge. Preferably, the actuator is disposed on, and extends outwardly from, an outer surface of the cartridge housing. In some embodiments, the actuator may be at least slightly tapered, such that insertion of the cartridge within an intravenous (IV) delivery system will depress the actuator when the cartridge is inserted into the delivery system.

(15) It is contemplated that the infusion pump could comprise one or more sensors to provide various information to the infusion pump, display, control unit, or other device. As one example, the pump unit or other component of the device could comprise one or more sensors which are configured to monitor a flow rate of the fluid within the tubing of the cartridge. As another example, the pump unit could comprise one or more sensors configured to sense or determine properties of the fluid within the tubing, such as a concentration or type of medication or other fluid within the tubing. Such sensors could include, for example, chemical sensors, optical sensors, resistance sensors, and so forth.

(16) Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

- (1) FIG. 1A illustrates a flowchart of one embodiment of a fluid delivery system.
- (2) FIG. 1B illustrates one embodiment of a fluid delivery system.
- (3) FIG. 2 illustrates one embodiment of an infusion pump and display.
- (4) FIG. 3 illustrates another embodiment of an infusion pump and display.
- (5) FIG. 4 illustrates another embodiment of an infusion pump and display.
- (6) FIG. 5 illustrates another embodiment of an infusion pump and display.
- (7) FIGS. 6A-6D illustrate one embodiment of a latching mechanism for an infusion pump.
- (8) FIGS. 7A-7B illustrate another embodiment of a latching mechanism for an infusion pump.
- (9) FIG. 8 illustrates one embodiment of a cartridge for an infusion pump.
- (10) FIG. 9A illustrates the cartridge of FIG. 8 with the bleed valve is a closed position.
- (11) FIG. 9B illustrates the cartridge of FIG. 8 with the bleed valve is an open position.
- (12) FIG. 10 illustrates another embodiment of a fluid delivery system.
- (13) FIG. 11 illustrates another embodiment of a fluid delivery system.
- (14) FIGS. 12A-12C illustrate the fluid delivery system of FIG. 11 illuminated in green, yellow and red, respectively.
- (15) FIGS. 13A-13B illustrate another embodiment of a pump unit for an infusion pump.
- (16) FIGS. 14A-14B illustrate another embodiment of a pump unit for an infusion pump.
- (17) FIG. 15A illustrates a cross-section view of another embodiment of an infusion pump.
- (18) FIG. 15B illustrates an enlarged portion of the infusion pump of FIG. 15A.
- (19) FIGS. 16A-16B illustrates a cross-section view of another embodiment of an infusion pump.
- (20) FIG. 16C illustrates an enlarged portion of the infusion pump of FIG. 16B.
- (21) FIG. 16D illustrates an exploded view of one embodiment of a piston and tie rod for a pump unit.
- (22) FIG. 16E illustrates an assembled view of the piston and tie rod of FIG. 16D.
- (23) FIGS. 16F-16G illustrate side views of one embodiment of an actuator and valve of a cartridge.

### **DETAILED DESCRIPTION**

(24) Throughout the following discussion, numerous references may be made regarding servers, services, interfaces, portals, platforms, or other systems formed from computing devices. It should be appreciated that the use of such terms is deemed to represent one or more computing devices having at least one processor configured to execute software instructions stored on a computer readable tangible, non-transitory medium. For example, a server can include one or more computers operating as a web server, database server, or other type of computer server in a manner to fulfill described roles, responsibilities, or functions.

(25) The following discussion provides many example embodiments of the inventive subject matter. Although each embodiment represents a single combination of inventive elements, the inventive subject matter is considered to include all possible combinations of the disclosed elements. Thus, if one embodiment comprises elements A, B, and C, and a second embodiment comprises elements B and D, then the inventive subject matter is also considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

(26) FIG. 1A illustrates a flowchart of one embodiment of a fluid delivery system **100**. System **100** preferably comprises an infusion pump **102** communicatively coupled with a display **104**, which is preferably separable from the pump **102**. The infusion pump **102** preferably receives information about one or more vital signs of a patient **112** from one or more sensors **106**. Such information can be received directly from the one or more sensors **106** or via an intermediary device such as a patient monitor **108**, for example.

(27) The infusion pump **102** preferably regulates a flow of medication **110** or other fluid through the pump **102** and thereby controls the flow rate of the fluid and thus a delivery of the medication **110** or other fluid to the patient **112**. The pump **102** preferably receives power from a line voltage **114**; however, it is contemplated that the pump **102** could have a battery backup to continue to regulate fluid flow if power to the infusion pump **102** is shut off or temporarily disconnected.

(28) In some embodiments, both the display **104** and the infusion pump **102** could be mounted to or otherwise coupled to a pole **118**.

(29) FIG. 1B illustrates one embodiment of a fluid delivery system, which comprises an infusion pump **102** communicatively coupled with a display **104**. Preferably, the display **104** is separable from the pump **102**. The infusion pump **102** preferably receives information about one or more vital signs of a patient **112** from one or more sensors, which could be monitored by a patient monitor **108** or another intermediary device. In other embodiments, the sensors could transmit information directly to the infusion pump **102**. Contemplated sensors could include, for example, a pressure transducer, a thermometer, a heart rate monitor, a blood oxygen monitor, and so forth. Using the display, for example, a medical professional **114** could input commands or review information about the patient **112** and/or medication delivery.

(30) The infusion pump **102** preferably regulates a flow of medication **110** or other fluid from a source such as an IV bag through the infusion pump **102** and to the patient **112**. Preferably, the infusion pump **102** controls a flow rate of the medication **110** or other fluid through the tubing **109** and thus controls an amount of the medication **110** or other fluid being delivered to the patient **112** over a specific period of time. In some embodiments, both the display **104** and the infusion pump **102** could be mounted to or otherwise coupled to a pole **118**.

(31) FIG. 2 illustrates one embodiment of an infusion pump **202** comprising a housing **204** configured to receive a cartridge **206** having a tubing **208**. The infusion pump **202** preferably comprises a controller having a processor and memory, wherein the memory is configured to store one or more control algorithms. The controller is configured to control a motor of the pump unit which thereby regulates a flow of the fluid within the tubing **208** of the cartridge **206**. In some embodiments, the controller can be configured to also control a latching mechanism such as one or more solenoids to release the cartridge **206**.

(32) A display **210** is communicatively coupled with the infusion pump **202** and configured to display information concerning a flow of medication to the patient as well as information concerning one or more vital signs of the patient. Preferably, the display **210** comprises a display screen which may be a touch screen or other technology that allows for inputting of one or more commands via touching or otherwise interacting with the display **210**. Of course, it is contemplated that commands could be inputted via other methods such as by voice, a keyboard, a mouse, or other input device. In this manner, it is contemplated that a command can be received at the display **210** and then transmitted from the display **210** to the infusion pump **202**.

(33) Preferably, each of the display **210** and the infusion pump **202** can be separable and

mechanically coupled to a pole **220**, such that both the display **210** and the infusion pump **202** can be mounted to the pole **220**. Preferred mounting apparatus allows for the display **210** and the infusion pump **202** to move vertically along the pole **220** to thereby adjust a height of each relative to the floor or other surface. In some embodiments, the mounting apparatus of the display **210** can also allow for angular adjustment such that the display **210** can be rotated and/or tilted with respect to the pole **220**.

(34) In some embodiments, the display **210** can be held in place via a dock **212**, which is mechanically coupled to the pole **220**. In this manner, the display **210** can be removably coupled to the dock **212**, such that the display **210** can be removed from the dock **212** and be held by a medical professional or other user, for example. It is contemplated that the dock **212** may comprise one or more connections that provide for power and a wired data connection to the display **210** when the display **210** is inserted into the dock **212**.

(35) Rather than include a control unit (controller) in the housing of the infusion pump **202**, it is alternatively contemplated that the infusion pump **302** can be communicatively coupled with a control unit **330** disposed outside of the housing, such as shown in FIG. 3. Similar to that discussed above, the control unit **330** preferably comprises a processor and memory, wherein the memory is configured to store one or more control algorithms. The control unit **330** is communicatively coupled with the infusion pump **302** (directly or indirectly via a wired or wireless connection) to control the infusion pump **302**, and in some embodiments, the control unit **330** may control multiple infusion pumps with each infusion pump regulating a flow of a different medication of fluid.

(36) The control unit **330** is configured to control a pump unit and/or motor of the infusion pump **302** which thereby regulates a flow of fluid within a tubing **308** of the cartridge **306** disposed within the pump **302**. In some embodiments, the control unit **330** can be configured to control a latching mechanism such as one or more solenoids to release the cartridge upon command of the controller. A more detailed discussion of exemplary latching mechanisms is provided below. With respect to the remaining numerals in FIG. 3, the same considerations for like components with like numerals of FIG. 2 apply.

(37) FIG. 4 illustrates a control unit **430** communicatively coupled to four infusion pumps **402A-402D**. Similar to that discussed above, the control unit **430** comprises a processor and memory, wherein the memory is configured to store one or more control algorithms. The control unit **430** is configured to control each of the infusion pumps **402A-402D**, and specifically, a motor of each of the infusion pumps **402A-402D** to regulate a flow of fluid within a tubing **408A-408D** of a cartridge **406A-406D** inserted within each of the infusion pumps **402A-402D**, respectively. In some embodiments, the control unit **430** can be configured to also control a latching mechanism such as one or more solenoids to release the cartridge **406A**, **406B**, **406C** or **406D** disposed within one of the infusion pumps **402A-402D**.

(38) The control units described herein with respect to the various embodiments and whether disposed within or outside of the infusion pump can be configured to receive signals and/or information and transmit command signals. For example, a control unit can be configured to receive signals and/or information concerning one or more vital signs of a patient, and based on the information received, change a dosage rate of medication to the patient. This can occur, for example, by the control unit causing the motor in a pump unit to increase or decrease the RPMs, such as by varying a voltage to the motor, based on one or more algorithms stored in a memory of the control unit and/or one or more variables, limits or factors.

(39) In contrast to the infusion pump shown in FIG. 2 in which the pump has an opening on the front-facing surface to receive a cartridge, the infusion pumps **402A-402D** each has an opening on a top surface of the infusion pump that is configured to receive a cartridge **406A-406D**, respectively. With respect to the remaining numerals in FIG. 4, the same considerations for like components with like numerals of FIG. 2 apply.

(40) FIG. 5 illustrates another embodiment of an infusion pump **502** comprising a housing **504** having a first opening **505** and a hollow interior portion. Preferably, a cartridge **506** can be inserted into the first opening **505** and at least partially within the hollow interior portion. Cartridge **506** preferably comprises a portion of a tubing **508**, which can be connected at one end to a fluid source **540** and at the other end to a patient. When the cartridge **506** is inserted within the housing **504**, the cartridge is preferably retained within the housing **504** using one or more latch mechanisms, such as described below.

(41) The infusion pump **502** can optionally include a handle **550** to facilitate carrying of the infusion pump **502**. In some embodiments, a power cable for the infusion pump **502** can be wrapped about the handle **550** when the cable is not in use. In other embodiments, the cable can be wrapped about a foot **552** of the infusion pump **502** when not in use.

(42) Display **510** can be communicatively coupled with the infusion pump **502** via a wired or wireless connection and configured to display information concerning a flow of medication to a patient as well as information concerning one or more vital signs of the patient on a display screen of the display **510**. Preferably, display **510** has a touch screen or other technology that allows for inputting of one or more commands via touching or otherwise interacting with the display **510**. In this manner, it is contemplated that a command can be received at the display **510** and then transmitted from the display **510** to the infusion pump **502**. In the embodiment shown in FIG. 5, it is contemplated that the infusion pump **502** comprises a control unit disposed within the housing **504** and configured to control a pump unit that is also disposed in the housing **504**. Exemplary pump units are described in more detail below.

(43) Preferably, each of the display **510** and the infusion pump **502** can be separable from one another and mechanically coupled to a pole **520**, such that both the display **510** and the infusion pump **502** can be mounted or otherwise coupled to the pole **520**. Preferred mounting apparatus allows for the display **510** and the infusion pump **502** to move vertically along the pole **520**. In some embodiments, the mounting apparatus of the display **510** can allow for angular adjustment such that the display **510** can be tilted and/or rotated with respect to the pole **520**. In such embodiments, the display **510** and infusion pump **502** can thereby be adjusted to a desired height for a medical professional **560**.

(44) In some embodiments, the display **510** can be held in place via a dock **512**, which is mechanically coupled to the pole **520**. In this manner, the display **510** can be removably coupled to the dock **512**, such that the display **510** can be removed and hand-held by a user, for example.

(45) Additionally or alternatively, it is contemplated that the display **510** could be hingedly coupled to the infusion pump **502** such that the display **510** could be rotated and/or tilted with respect to the infusion pump **502**. In alternative embodiments, it is contemplated that the infusion pump **502** could have a slot or other mechanism to receive a portion of the display **510** such that the display **510** can be secured to the pump **502** during transport, for example.

(46) With respect to the remaining numerals in FIG. 5, the same considerations for like components with like numerals of FIG. 2 apply.

(47) FIGS. 6A-6D illustrate one embodiment of a latching mechanism **640** comprising a latching solenoid that is configured to engage an aperture or notch/recess/indentation **607** of the cartridge **606** having a tubing **608** when the cartridge **606** is properly situated within the housing **604** of the infusion pump **602**. Preferably, the latching mechanism **640** is a pull solenoid that is biased in a first position shown in FIGS. 6A-6B, in which a projection of the latching mechanism **640** is disposed within the aperture or notch/recess/indentation **607** of the cartridge **606**. It is preferred that the cartridge **606** has a curved or tapered upper surface and, optionally a curved or tapered bottom surface, such that the latching mechanism **640** is depressed as the cartridge **606** is inserted and until the latching mechanism **640** is displaced into the aperture or notch/recess/indentation **607** of an outer surface of the cartridge **606**.

(48) As shown in FIGS. 6A-6D, the latching mechanism **640** can be mechanically coupled to an



emergency release **642** that can be hidden by a cover **644** on an outer surface of the housing **604** when not in use. The emergency release **642** can advantageously be used to change a position of the latching mechanism **640** to the position shown in FIG. **6C**, and thereby release the cartridge **606** when the latching mechanism **640** is not functioning such as due to a power outage or failure of the latching mechanism **640**. In some embodiments, the emergency release **642** comprises a metal part with a snap ring that when pulled releases the engagement of the latching mechanism **640** with the cartridge **606**, such as shown in FIG. **6C**.

(49) FIGS. **7A-7B** illustrate another embodiment of infusion pump **702** having a housing **704** into which a cartridge **706** may be inserted. As discussed above, the pump **702** can include a motor **750** that is connected to and drives a pump assembly **752**, which is used to cause and regulate a flow of fluid within a tubing **708** of the cartridge **706**.

(50) The infusion pump **702** may comprise one or more latching mechanisms **740** configured to keep the cartridge **706** from inadvertently being removed from the pump **702**. Preferred latching mechanisms **740** comprise a latching solenoid that is configured to engage a recess or notch **707** of the cartridge **706** when the cartridge **706** is properly disposed within the housing **704** of the infusion pump **702**. Preferably, the latching mechanism **740** is a pull solenoid that is biased in a first position (shown in FIG. **7A**) where a projection of the latching mechanism **740** engages notch **707** in the sidewall of the cartridge **706**. As shown in FIG. **7A-7B**, preferably the housing **704** of the infusion pump **702** comprises two latching mechanisms **740** disposed on left and right sides of the housing **704**. In this manner, when the cartridge **706** is properly inserted within the housing **704**, the latching mechanisms **740** each engage a recess or notch **707** on the left and right sides of the cartridge **706**.

(51) As discussed above in relation to the latching mechanism of FIGS. **6A-6D**, it is contemplated that latching mechanisms **740** could each have an emergency release such as that described above to ensure the cartridge **706** can be manually released from the housing **704** if needed.

(52) FIG. **8** illustrates one embodiment of a cartridge **806** having a cartridge housing **809** and a tubing **808**, with a portion of the tubing **808** extending through the cartridge **806** to create a C-shape. As shown, cartridge housing **809** can comprise notches or recessed areas **807** on left and right sides of the cartridge housing **809** that are each configured to receive a portion of a latching mechanism, such as those described herein.

(53) In some embodiments, the left and right sides of the cartridge housing **809** may comprise a recessed area **811** that leads to the notch or recessed area **807**, which thereby keeps the latching mechanism disengaged until the latching mechanism is at the notch or recessed area **807**.

(54) As shown in FIG. **8**, the tubing is preferably held flat at the back of the cartridge **806**, such that when the cartridge **806** is properly inserted within the housing of the infusion device, the tubing at the back of the cartridge **806** is pressed against a pump unit. Tubing **808** preferably enters and exits through a front surface of the cartridge **806** and loops about to form a C-shape.

(55) Cartridge **806** further comprises a valve **820** that facilitates priming of the cartridge **806**. The valve **820** can comprise a bleed valve, which can be actuated using a push button or pull tab, for example. When the valve **820** is in a first position, the tubing **808** can be compressed to thereby prevent fluid from flowing through the tubing. When the valve **820** is in a second position, the tubing **808** is decompressed such that fluid can flow through the tubing **808**, such as to prime the tubing.

(56) In some embodiments, the cartridge housing **809** is preferably transparent or translucent, to allow for light to be emitted through the cartridge **806**.

(57) FIG. **9A** illustrates cartridge **806** in which the valve **820** is in the first position (tubing compressed). FIG. **9B** illustrates the cartridge **806** in which the valve **820** is in the second position (tubing decompressed).

(58) In an alternative embodiment, an actuator coupled to a valve could be used, where the actuator comprises a push button to cause the valve to move from a first position (closed) to a second

position (open) when the actuator is depressed. It is contemplated that the valve could comprise a hook that is disposed about the tubing such as described below and shown in FIGS. 16F-16G. The valve could have a spring that causes the valve to be biased in the first position where the hook compresses the tubing and prevents fluid flow through the tubing. When the actuator is pressed, the hook can move away from the tubing allow fluid to flow through the tubing. Preferably, the push button actuator is disposed on, and extends outwardly away from, an outer surface of the cartridge housing, such as the top surface. Because the cartridge is sized to be held in one hand, the user can prime the cartridge by depressing the push button actuator with a finger of the hand holding the cartridge. In such embodiments, it is contemplated that the push button actuator can be depressed when the cartridge is inserted within the housing of the infusion pump such that fluid can flow through the tubing.

(59) FIG. 10 illustrates a fluid delivery system **1000** of an infusion pump that comprises a pump unit **1002** disposed within the housing of an infusion pump. Preferably the infusion pump is configured to receive a cartridge **1004** having a tubing **1006** that extends through the cartridge **1004** to form a C-shape, for example. When the cartridge **1004** is inserted into the housing of the infusion pump, the tubing **1006** at the back of the cartridge **1004** is pressed against a portion of the pump unit **1002**.

(60) As shown, the pump unit **1002** comprises a variable-speed motor **1008** mechanically coupled to an eccentric cam **1012** via a set of gears **1010**. In this manner, motor **1008** causes rotation of the cam **1012**. Preferably, the cam **1012** comprises a single injection molded piece to reduce overall wear of the cam **1012** over time. The cam **1012** preferably comprises a plurality of circular disks or sheaves **1013** disposed on an axle of the cam **1012**, with at least some of the disks or sheaves **1013** being disposed on the axle such that a center of the disk or sheave **1013** is offset from a center of the axle of the cam **1012**. It is contemplated that the cam **1012** could be modular and replaceable, which allows the pump unit **1002** to be easily serviced over the lifespan of the pump unit **1002**.

(61) The cam **1012** is preferably adjacent to one or more pads **1014**, which are pressed sequentially against the tubing **1006** as the cam **1012** rotates and the sheaves **1013** contact the pads **1014**. As the pads **1014** are pressed against the tubing **1006** and then released, this causes fluid to flow within the tubing **1006**. The more often the pads **1014** press against the tubing **1006** and then are released within a predetermined period of time, the greater the flow rate of the fluid through the tubing.

(62) The fluid delivery system **1000** preferably comprises latching mechanisms **1020**, which retain the cartridge **1004** within the housing of the infusion pump when engaged with the cartridge **1004**. Preferably, the latching mechanisms **1020** comprise pull solenoids that engage with a notch or recessed area on left and right sides of the cartridge **1004**, such as described above.

(63) In some embodiments, cartridge **1004** comprises left and right indentations to facilitate a user's grip on the cartridge **1004**.

(64) FIG. 11 illustrates another embodiment of a fluid delivery system **1100** of an infusion pump having light sources **1130** that can be used to illuminate the cartridge **1104**, for example. In this manner, it is contemplated that the light sources **1130** can be used to visually indicate, for example, a status of the fluid delivery system **1100**, when an alert is generated, when an alarm is generated, when the system requires attention, and so forth.

(65) FIGS. 12A-12C illustrate the cartridge being illuminated by light sources within the fluid delivery system **1100**, with FIG. 12A showing the cartridge being illuminated green, FIG. 12B showing the cartridge being illuminated yellow, and FIG. 12C showing the cartridge being illuminated red.

(66) FIG. 13A-13B illustrate an alternative embodiment of a pump unit **1302** having a crank shaft **1312** to which pistons **1320** are attached via connecting rods **1318**. As shown in FIG. 13A, the connecting rods **1318** are mounted eccentrically to an axle of the crank shaft **1312**. Like the pump unit described above with respect to FIG. 10, the crank shaft **1312** can be driven by a variable-speed motor mechanically coupled to the crank shaft **1312** via one or more gears. In this manner,

the motor can cause rotation of the crank shaft **1312** and control a speed of rotation of the crank shaft **1312**.

(67) Preferably, the crank shaft **1312** comprises a single injection molded piece to reduce overall wear of the crank shaft **1312** over time. The crank shaft **1312** preferably comprises a plurality of disks or sheaves **1316** disposed on an axle of the crank shaft **1312**, with adjacent pairs of the disks or sheaves **1316** connected by a connecting bar **1314** that is disposed off-center of an axis of the crank shaft **1312**. One end of each of the connecting rods **1318** is connected to one of the connecting bars **1314**.

(68) As the crank shaft **1312** rotates, the disks or sheaves **1316** also rotate which cause the connected pistons **1320** to move over time. It is contemplated that the pistons **1320** can be configured to contact the tubing **1332** of the cartridge **1330** as they move with respect to the axle of the crank shaft **1312** to thereby cause fluid to flow within the tubing **1332**, such as shown in FIG. **13B**. The pistons **1320** move in a specific manner with respect to the tubing **1332** to thereby force a flow of fluid within the tubing **1332**. An increase in a rate of movement of the pistons **1320** will similarly increase a flow rate of the fluid within the tubing **1332**. Similarly, a decrease in a rate of movement of the pistons **1320** will similarly decrease a flow rate of the fluid within the tubing **1332**. With respect to the remaining numerals in FIGS. **13A-13B**, the same considerations for like components with like numerals of FIG. **10** apply.

(69) FIG. **14A-14B** illustrate an alternative embodiment of a pump unit **1402** of an infusion pump having an eccentric cam **1412** comprising a plurality of circular disks or sheaves **1416** disposed on an axle of the cam **1412** such that a center of the disk or sheave **1416** is offset from a center of the axle of the cam **1412**. Like the pump unit described above with respect to FIG. **10**, the cam **1412** can be driven by a variable-speed motor mechanically coupled to the cam **1412** via one or more gears. In this manner, the motor causes rotation of the cam **1412**.

(70) Preferably, the cam **1412** comprises a single injection molded piece to reduce overall wear of the cam **1412** over time.

(71) A plurality of intermediary pieces **1424** are disposed between the tubing **1432** of the cartridge **1430** and the disks or sheaves **1416**, such that each intermediary piece **1424** is adjacent to one of the disks or sheaves **1416** and the tubing **1432**. Each of the intermediary pieces **1424** can be pivotally coupled to a stationary piece **1426** via a connecting rod **1414**. In this manner, as the cam **1412** rotates, each of the disks or sheaves **1416** will cause different ones of the intermediary pieces **1424** to press against the tubing **1432**, such as shown in FIG. **14A**, and then move away from the tubing **1432**, such as shown in FIG. **14B**. This generally sequential movement of the intermediary pieces **1424** against the tubing **1432** causes fluid to flow within the tubing at a rate corresponding to a rotation of the cam **1412**.

(72) FIGS. **15A-15B** illustrates cross-sectional views of another embodiment of an infusion pump **1502** comprising a housing **1504** configured to receive a replaceable and/or disposable cartridge **1506** having a tubing **1508**. As discussed above, when the cartridge **1506** is properly inserted into the infusion pump **1502**, a portion of the tubing **1508** of the cartridge **1506** is disposed adjacent to a pump unit **1562**. The tubing **1508** may extend through the cartridge **1506** to form a C-shape, for example.

(73) The infusion pump **1502** preferably comprises a controller **1550** having a processor and memory, wherein the memory is configured to store one or more control algorithms. The controller **1550** is configured to control a motor **1560** of the pump unit **1562** which thereby regulates a flow of the fluid within the tubing **1508** of the cartridge **1506**. In some embodiments, the controller **1550** can be configured to also control a latching mechanism **1540** such as one or more solenoids to release the cartridge **1506**. The latching mechanisms **1540** can retain the cartridge **1506** within the housing **1504** of the infusion pump **1502** when engaged with the cartridge **1506**. Preferably, the latching mechanisms **1540** comprise pull solenoids that engage with a notch or recessed area on left and right sides of the cartridge **1506**, such as described above. In some embodiments, the cartridge

**1506** comprises left and right indentations to facilitate a user's grip on the cartridge **1506**.

(74) Preferably, the motor **1560** comprises a variable-speed motor mechanically coupled to a crank shaft **1564** via a set of gears **1566**. In this manner, the motor **1560** causes rotation of the crank shaft **1564** and control a speed of rotation of the crank shaft **1564**. Preferably, the crank shaft **1564** comprises a single injection molded piece to reduce overall wear of the crank shaft **1564** over time. The crank shaft **1564** preferably comprises a plurality of circular disks or sheaves **1568** disposed on an axle of the crank shaft **1564**. In some embodiments, at least some of the disks or sheaves **1568** are disposed on the axle such that a center of at least some of the disks or sheaves **1568** is offset from a center of the axle of the crank shaft **1564**. It is contemplated that the crank shaft **1564** could be modular and replaceable, which allows the pump unit **1562** to be easily serviced over the lifespan of the pump unit **1562**.

(75) A set of pistons **1570** are each coupled to the crank shaft **1564** via a tie rod **1569**. In preferred embodiments, the tie rods **1569** are mounted eccentrically to the axle of the crank shaft **1564**. In some embodiments, the crank shaft **1564** the disks or sheaves **1568** are disposed on the axle of the crank shaft **1564**, with adjacent pairs of the disks or sheaves **1568** connected by a connecting bar that is disposed off-center of the axis of the crank shaft **1564**. One end of each of the tie rods **1569** is connected to one of the connecting bars, such as described above.

(76) As the crank shaft **1564** rotates, the disks or sheaves **1568** also rotate which cause the connected pistons **1570** to move over time (compare position of pistons **1570** in FIG. 15A with FIG. 15B). It is contemplated that the pistons **1570** can be configured to contact the tubing **1508** of the cartridge **1506** as the pistons **1570** move back and forth with respect to the axle of the crank shaft **1564** to thereby cause fluid to flow within the tubing **1508** of the cartridge **1506**. The pistons **1570** move in a specific manner with respect to the tubing **1508** to thereby force a flow of fluid within the tubing **1508**. An increase in a rate of movement of the pistons **1570** will similarly increase a flow rate of the fluid within the tubing **1508**. Similarly, a decrease in a rate of movement of the pistons **1570** will similarly decrease a flow rate of the fluid within the tubing **1508**.

(77) FIGS. 16A-16G illustrates cross-sectional views of another embodiment of an infusion pump **1602** comprising a housing **1604** configured to receive a replaceable and/or disposable cartridge **1606** having a tubing **1608**. When the cartridge **1606** is properly inserted into the infusion pump **1602**, as shown in FIG. 16B, a portion of the tubing **1608** of the cartridge **1606** is disposed adjacent to pistons **1670** of a pump unit **1662**. The tubing **1608** may extend through the cartridge **1606** to form a C-shape, for example.

(78) The infusion pump may also comprise one or more latching mechanisms **1640** that can be used to secure and release the cartridge **1606**. For example, the latching mechanisms **1640** can retain the cartridge **1606** within the housing **1604** of the infusion pump **1602** when engaged with the cartridge **1606**. Preferably, the latching mechanisms **1640** comprise pull solenoids that engage with a notch or recessed area on left and right sides of the cartridge **1606**, such as described above. In some embodiments, the cartridge **1606** comprises left and right indentations to facilitate a user's grip on the cartridge **1606**.

(79) The infusion pump **1602** preferably comprises a motor **1660** configured to control a pump unit **1662** and thereby regulate a flow of the fluid within the tubing **1608** of the cartridge **1606**. In some embodiments, a controller (such as shown in FIG. 15A) can be configured to control a speed of the motor **1660**. Preferably, the motor **1660** comprises a variable-speed motor mechanically coupled to a crank shaft **1664** via a set of gears **1666**.

(80) FIG. 16C illustrates an enlarged portion of FIG. 16B, showing how the pistons **1670** interact with the tubing **1608** of the cartridge **1606**. As can be seen, the motor causes rotation of the gears **1666**, which in turn rotate the crank shaft **1664**. As the crank shaft rotates, different ones of the pistons **1670** are pressed against the tubing **1608** causing the tubing **1608** to be compressed, while others of the pistons **1670** are moved away from the tubing **1608** and allowing the tubing **1608** to fill with fluid. The sequential nature of the movement of the pistons **1670** causes the fluid to move

through the tubing **1608** in the direction shown in the Figure. An increase in a rate of movement of the pistons **1670** will similarly increase a flow rate of the fluid within the tubing **1608**. Similarly, a decrease in a rate of movement of the pistons **1670** will similarly decrease a flow rate of the fluid within the tubing **1608**.

(81) FIG. **16D** illustrates an exploded view of a piston **1670** and a tie rod **1669**, while FIG. **16E** illustrates an assembled view of that portion of the pump unit **1662**. As shown, the piston **1670** is connected to a circular disk or sheave **1668** by the tie rod **1669**, with one end of the tie rod **1669** having an aperture **1671** that is configured to receive a projection **1672** of the circular disk or sheave **1668**.

(82) As shown in FIG. **16C**, the crank shaft **1664** preferably comprises a plurality of the circular disks or sheaves **1668** disposed in series along on an axle of the crank shaft **1664**. In preferred embodiments, the tie rods **1669** are mounted eccentrically to the axle of the crank shaft **1664**. For example, the projection **1672** of each sheave **1668** may be disposed off center from a center of the sheave **1668** and/or the sheave **1668** may be an ovular or other shape. It is contemplated that the sheaves **1668** are sized and shaped and/or disposed on the axle of the crank shaft **1664** in such a manner so that different ones of the pistons **1670** will press against the tubing **1608** at different times based on the relationship between the piston **1670**, the sheave **1668** and the crank shaft **1664**.

(83) In preferred embodiments, the cartridge **1606** comprises a valve **1680** that can be used to inhibit a flow of fluid within the tubing **1608** of the cartridge **1606**. Preferably, the valve **1680** is biased in a closed position shown in FIG. **16A** and FIG. **16F** such as by use of a spring **1681** that causes the valve to compress a stopper **1682** against the tubing **1608**. The valve preferably comprises the stopper **1682** and a hook that wraps around at least a portion of the tubing **1608**. When the stopper **1682** and hook are moved closer together, this compresses the tubing **1608** preventing fluid from flowing through the compressed portion of the tubing **1608** as shown in FIG. **16F**.

(84) An actuator **1684** can be used to move the hook of the valve **1680** away from the stopper **1682** and thereby no longer compress the tubing **1608**. Thus, when a force is applied to the actuator **1684**, such as shown in FIG. **16G**, the tubing **1608** is no longer compressed and fluid can flow through the tubing **1608**.

(85) Preferably, the actuator **1684** is disposed on, and extends outwardly away from, an outer surface of the cartridge **1606**. Because the cartridge **1606** is sized to be held in one hand, the medical professional can hold the cartridge in one hand and prime the cartridge **1606** by depressing the actuator **1684** using a finger of that. As shown in FIG. **16B**, the actuator **1684** can be depressed and remain in that position by a surface of the housing **1604** of the pump **1602** when the cartridge **1606** is inserted within the housing **1604**.

(86) The various embodiments described above can further including a spring or other mechanism positioned near the back of the cartridge near the pump unit that facilitates ejection of the cartridge from the infusion pump when the latching mechanism is released. Rather than, or in addition to, a spring, a push solenoid could be disposed within the housing and configured to contact the cartridge and facilitate ejection of the cartridge from the hollow interior portion when the push solenoid is actuated.

(87) As used herein, and unless the context dictates otherwise, the term “coupled to” is intended to include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements). Therefore, the terms “coupled to” and “coupled with” are used synonymously.

(88) In some embodiments, the numbers expressing measurements, quantities of ingredients, properties such as concentration, reaction conditions, and so forth, used to describe and claim certain embodiments of the invention are to be understood as being modified in some instances by the term “about.” Accordingly, in some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the

desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the invention may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

(89) Unless the context dictates the contrary, all ranges set forth herein should be interpreted as being inclusive of their endpoints and open-ended ranges should be interpreted to include only commercially practical values. Similarly, all lists of values should be considered as inclusive of intermediate values unless the context indicates the contrary.

(90) As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(91) The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value with a range is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

(92) Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

(93) It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refers to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

## Claims

1. A fluid delivery system, comprising: a pump unit disposed within the housing and comprising a motor mechanically coupled with a crank shaft having a set of sheaves disposed on or connected to the crank shaft; wherein the pump unit further comprises a set of pistons and a set of tie rods, and wherein each of the pistons is coupled to the crank shaft by one of the tie rods, such that the tie rod can move with respect to the piston, and wherein each of the tie rods are connected to one of the

- sheaves, such that the tie rod can move with respect to the sheave; and wherein the pistons are configured to move over time as the crank shaft rotates.
2. The fluid delivery system of claim 1, wherein the crank shaft and the set of sheaves comprise a single injection molded piece.
  3. The fluid delivery system of claim 1, wherein the set of sheaves each comprises a projection that is configured to be received by one of the tie rods which connects the tie rod to the sheave, and wherein the projection is disposed off center from an axle of the crank shaft and from a center of the sheave.
  4. The fluid delivery system of claim 1, further comprising: a controller having a processor and memory and configured to receive information and transmit command signals; and wherein the controller is communicatively coupled to the pump unit and wherein the pump unit is configured to receive the command signals to thereby control a speed of the motor.
  5. The fluid delivery system of claim 4, further comprising: an infusion pump having a housing with a first opening and a hollow interior portion that is configured to receive a cartridge having the tubing, wherein when the cartridge is disposed within the hollow interior portion, the pistons are configured to compress different portions of the tubing of the cartridge over time as the crank shaft rotates; a second infusion pump having a second housing with a second opening and a second hollow interior portion that is configured to receive a second cartridge having a second tubing; a second pump unit disposed within the housing and comprising a second motor mechanically coupled with a second crank shaft having a second set of sheaves disposed on or connected to the second crank shaft; and wherein the second pump unit further comprises a second set of pistons and a second set of tie rods, and wherein each of the pistons of the second set is coupled to the second crank shaft by one of the tie rods of the second set, and wherein each of the tie rods of the second set are connected to one of the sheaves of the second set; wherein when the second cartridge is disposed within the second hollow interior portion, the pistons of the second set are configured to compress different portions of the second tubing of the second cartridge over time as the second crank shaft rotates; and wherein the controller is communicatively coupled to the second pump unit and wherein the second pump unit is configured to receive the command signals.
  6. The fluid delivery system of claim 1, further comprising: at least one light source disposed within the housing and configured to illuminate the hollow interior and the cartridge.
  7. The fluid delivery system of claim 1, further comprising: a display communicatively coupled with an infusion pump comprising the pump unit, wherein the display can receive information from the infusion pump and transmit commands to the infusion pump to control the pump unit.
  8. The fluid delivery system of claim 7, wherein the infusion pump further comprises a housing that comprises a slot configured to receive a portion of the display.
  9. The fluid delivery system of claim 1, further comprising a sensor configured to monitor a flow rate of fluid within the tubing.
  10. The fluid delivery system of claim 1, further comprising a sensor configured to determine properties of fluid within the tubing.
  11. A cartridge for a fluid delivery system, comprising: a housing comprising a top portion, a side portion, and a bottom portion, wherein the housing is configured to retain a portion of a tubing through which a medication can flow from a medication source to a patient, and wherein the tubing extends through the cartridge along the top portion, the side portion, and the bottom portion to form a C-shape, and wherein the tubing extends outside of the cartridge along at least one of the top portion, the side portion, and the bottom portion; a valve configured to compress the tubing and prevent fluid flow in the tubing when the valve is in a first position; and an actuator mechanically coupled to the valve, such that when a force is applied to the actuator, the valve is moved to a second position that allows fluid to flow through the tubing; and wherein the valve is biased in the first position.
  12. The cartridge of claim 11, wherein the valve comprises a hook and a stopper, and wherein the

tubing is disposed between the stopper and the hook, and wherein the hook is configured to compress the tubing against the stopper and prevent fluid flow in the tubing when the valve is in the first position.

13. The cartridge of claim 11, wherein the housing comprises tapered top and bottom surfaces.

14. The cartridge of claim 11, wherein the housing is translucent.

15. The cartridge of claim 11, wherein the actuator is disposed on the top portion of the housing.

16. A fluid delivery system, comprising: a cartridge comprising: a cartridge housing comprising a top portion, a side portion, and a bottom portion, wherein the cartridge housing is configured to retain a portion of a tubing through which a medication can flow from a medication source to a patient, wherein the tubing extends through the cartridge along the side portion; an infusion pump having a housing with a first opening and a hollow interior portion that is configured to receive the cartridge; a pump unit disposed within the housing and comprising a motor mechanically coupled with a crank shaft having a set of sheaves disposed on or connected to the crank shaft; wherein the pump unit further comprises a set of pistons and a set of tie rods, and wherein each of the pistons is coupled to the crank shaft by one of the tie rods, such that the tie rod can move with respect to the piston, and wherein each of the tie rods is coupled to one of the sheaves, such that the tie rod can move with respect to the sheave; and wherein when the cartridge is disposed within the hollow interior portion, at least a portion of the tubing that extends along the side portion of the cartridge is compressed by one or more of the pistons over time as the crank shaft rotates.

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